

DW_arb_sp

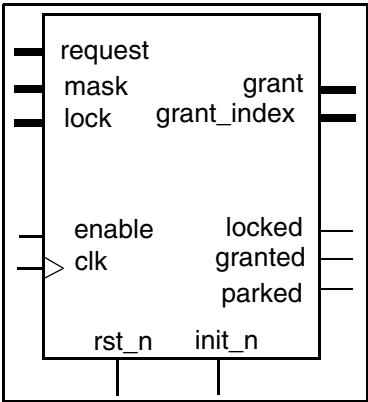
Arbiter with Static Priority Scheme

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterizable number of clients
- Programmable mask for all clients
- Park feature - default grant to a client when no requests are pending
- Lock feature - ability to lock the currently granted client
- Registered/unregistered outputs
- Includes a low-power implementation that has power benefits from minPower optimization (for details, see [Table 1-3](#) on page 2)

Revision History



Applications

- Control application
- Networking
- Bus interfaces

Description

DW_arb_sp implements a parametrized, synchronous arbiter with a static fixed priority scheme.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
clk	1 bit	Input	Input clock
rst_n	1 bit	Input	Asynchronous reset for all registers (active low)
init_n	1 bit	Input	Synchronous reset for all registers (active low)
enable	1 bit	Input	Enables clocking (active high)
request	<i>n</i> bits	Input	Input request from clients
lock	<i>n</i> bits	Input	Active high signal to lock input ■ For lock(<i>i</i>) = 1, the arbiter is locked to the request (<i>i</i>) if it is currently granted. ■ For lock(<i>i</i>) = 0, the lock on the arbiter is removed.

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
mask	n bits	Input	Active high input to mask specific clients - For $\text{mask}(i) = 1$, $\text{request}(i)$ is masked. - For $\text{mask}(i) = 0$, the mask on the $\text{request}(i)$ is removed.
parked	1 bit	Output	Flag to indicate that there are no requesting clients and the grant of resources has defaulted to <i>park_index</i>
granted	1 bit	Output	Flag to indicate that arbiter has issued a grant to one of the clients
locked	1 bit	Output	Flags that the arbiter is locked by a client
grant	n bits	Output	Grant output
grant_index	$\log_2 n$ bits	Output	Index of the requesting client that has been currently issued the grant or the client designated by <i>park_index</i> in <i>park_mode</i>

Table 1-2 Parameter Description

Parameter	Values	Description
n	2 to 32 Default: 4	Number of arbiter clients
park_mode	0 or 1 Default: 1	1: Includes logic to enable parking when no clients are requesting 0: Contains no logic for parking
park_index	0 to $n - 1$ Default: 0	Index of the client used for parking
output_mode	0 or 1 Default: 1	1: Includes registers at the outputs (see Figure 1-1 on page 4) 0: Contains no output registers (see Figure 1-2 on page 4)

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	RTL synthesis model	DesignWare
lpwr ^a	Low Power synthesis model	- DesignWare (P-2019.03 and later) - DesignWare-LP (before P-2019.03)

a. Requires that you enable minPower; for details, see [“Enabling minPower”](#) on page 7.
When minPower is enabled, the lpwr implementation is always chosen during synthesis.

Table 1-4 Simulation Models

Model	Function
DW05.DW_ARB_SP_SIM_CFG	Design unit name for VHDL simulation
dw/dw05/DW_arb_sp_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_arb_sp.v	Verilog simulation model source code

Table 1-5 Arbiter Status Flags

Flag	Characteristic	Description
parked	If <code>parked</code> is active, there are no active requests at the input of the arbiter.	The <code>parked</code> output, active high, indicates that grant of the resources has defaulted to the client defined by <code>park_index</code> in <code>park_mode = 1</code> . In <code>park_mode = 0</code> , this flag does not exist.
granted	If <code>granted</code> is active, there is at least one active request at the input of the arbiter.	The <code>granted</code> output, active high, indicates that the grant of resources is to one of the actively requesting inputs.
locked	If <code>locked</code> is active, the current <code>grant</code> and the corresponding <code>lock</code> signal must be active.	The <code>locked</code> output, active high, indicates that the currently granted client has locked out all other clients.

With the static fixed priority scheme, each client connected to the arbiter is assigned a unique priority. The order of priorities of these clients is indicated in the index of the input request port to which they are connected. For example, the client connected to the input `request[0]` has the highest priority, and the priorities of other clients descend correspondingly to the lowest priority for the client connected to `request[n-1]`.

The lock feature enables a client, despite requests from other clients, to have an exclusive grant for the duration of the corresponding lock input. After a client receives the grant, it can lock out other clients from the arbitration process by setting the corresponding `lock` input.

The park feature allows the resources to be granted to a designated client defined by the `park_index` parameter when there are no active requests pending. The `park_mode` parameter enables/disables the parking feature.

By setting the desired bits of the `mask` input, the corresponding clients can be masked off from consideration for arbitration. The mask on a client remains active until the corresponding `mask` input for the client is reset.

All the input requests from the arbiter clients are assumed to be synchronized to the arbiter clock signal `clk`. The arbiter provides flags `locked`, `granted`, and `parked`, to indicate the status of the arbiter.

As described in Table 1-2 on page 2, the DW_arb_sp can handle up to 32 requesting clients. The `mask`, `park` and `lock` features add flexibility to the arbiter. The parking of grant to a designated client saves an arbitration cycle and the parked client can lock the grant without issuing a request to the arbiter.

Figure 1-1 Block Diagram of DW_arb_sp Arbiter, output_mode = 1

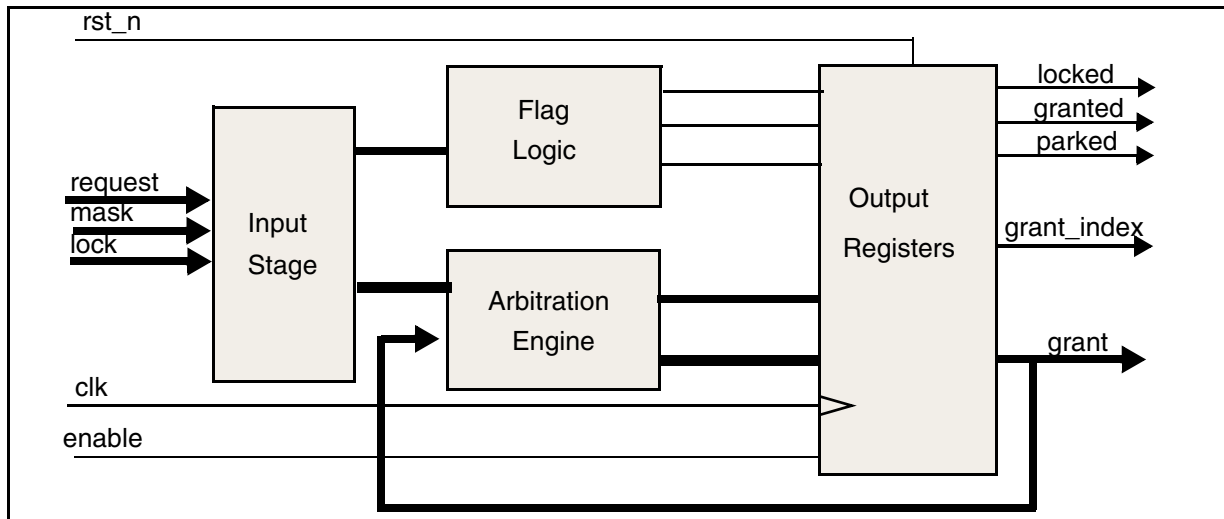
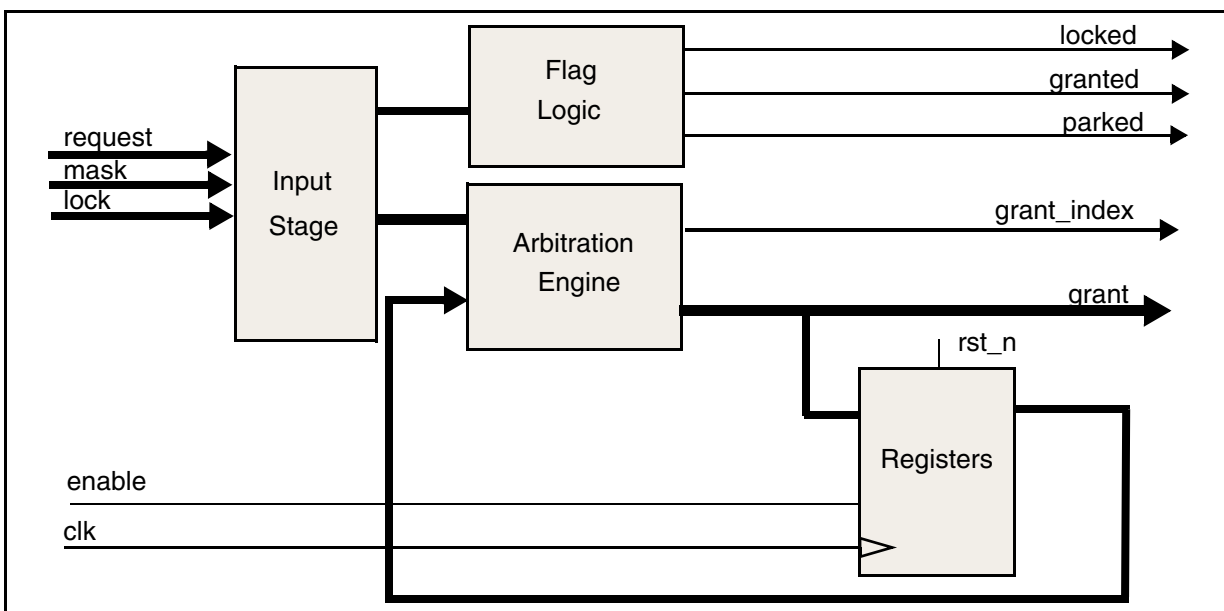


Figure 1-2 Block Diagram of DW_arb_sp Arbiter output_mode = 0



Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW_DISABLE_CLK_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

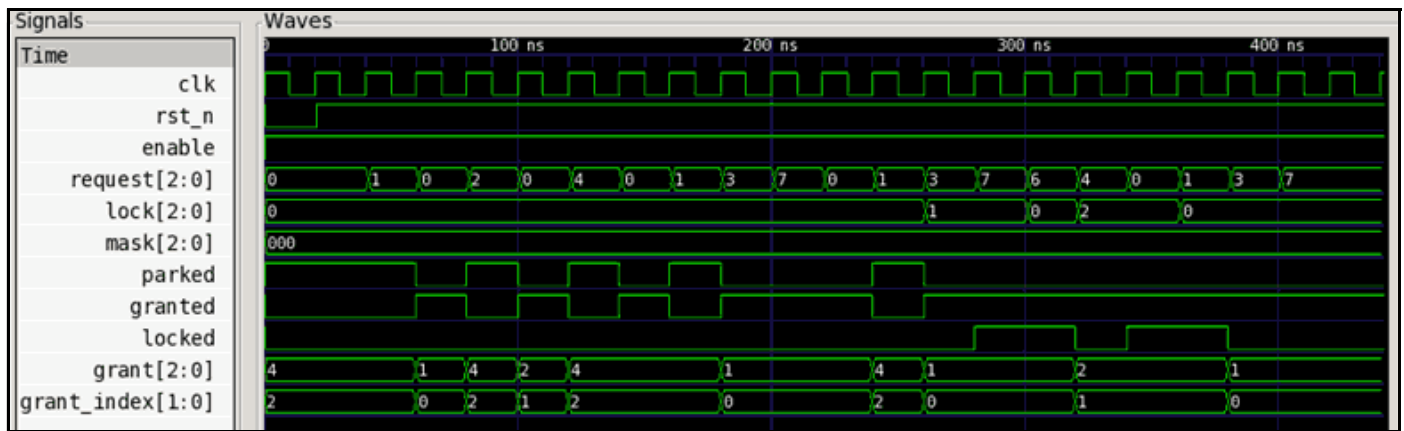
- Or, include a command line option to the simulator, such as:

```
+define+DW_DISABLE_CLK_MONITOR (which is used for the Synopsys VCS simulator)
```

This message is also suppressed using the DW_SUPPRESS_WARN macro explained earlier.

Timing Waveforms

Figure 1-3 Waveform 1: $n = 3$, $park_mode = 1$, $park_index = 2$ with $mask = 0$, $output_mode = 1$



Any client can be masked off by setting the corresponding mask bit. By doing so it will not be considered for the arbitration. If mask bits are set and none of the non-masked clients are actively requesting, the arbiter will be parked to the designated client defined by `park_index`. In the non-locked state of the arbiter, setting the mask bit of the currently granted client effectively invalidates the request from the client. In the following cycle, the current grant is de-asserted, and based on the current unmasked requests from other clients, a new client is generated. However, when a client has locked the arbiter, setting the mask bit of any client has no effect on the current grant.

Figure 1-4 Waveform 2: $n = 3$, $park_mode = 1$, $park_index = 2$, $mask = 4$ and $output_mode = 1$

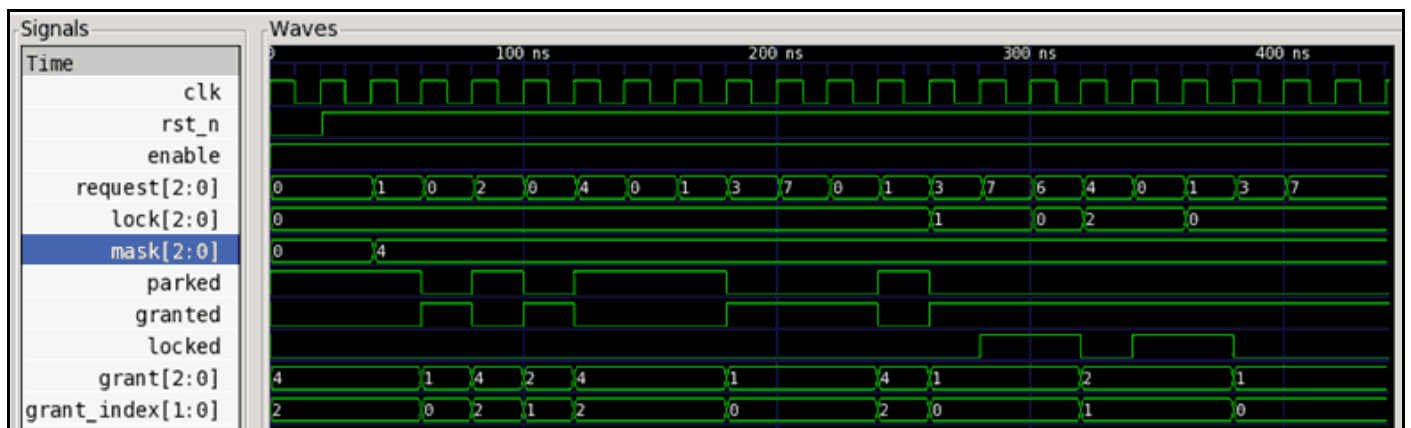
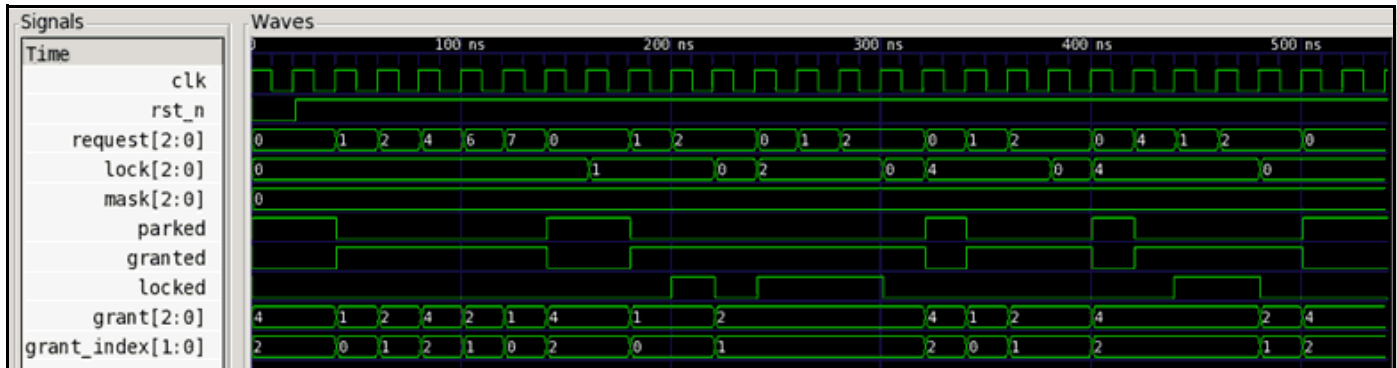


Figure 1-5 Waveform 3: $n = 3$, $park_mode = 1$, $park_index = 2$, $mask = 0$ and $output_mode = 0$ 

Enabling minPower

You can instantiate this component without enabling minPower, but to achieve power savings from the low-power implementation “lpwr” (see [Table 1-3](#) on page 2), you must enable minPower optimization, as follows:

- Design Compiler

- Version P-2019.03 and later:

```
set power_enable_minpower true
```

- Before version P-2019.03 (requires the DesignWare-LP license feature):

```
set synthetic_library {dw_foundation.sldb dw_minpower.sldb}
set link_library {* $target_library $synthetic_library}
```

- Fusion Compiler

Optimization for minPower is enabled as part of the total_power metric setting. To enable the total_power metric, use the following:

```
set_qor_strategy -stage synthesis -metric total_power
```

Related Topics

- [Application Specific – Control Logic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.dw_foundation_comp.all;

entity DW_arb_sp_inst is
  generic (
    inst_n : NATURAL := 4;
    inst_park_mode : NATURAL := 1;
    inst_park_index : NATURAL := 0;
    inst_output_mode : NATURAL := 1
  );
  port (
    inst_clk : in std_logic;
    inst_rst_n : in std_logic;
    inst_init_n : in std_logic;
    inst_enable : in std_logic;
    inst_request : in std_logic_vector(inst_n-1 downto 0);
    inst_lock : in std_logic_vector(inst_n-1 downto 0);
    inst_mask : in std_logic_vector(inst_n-1 downto 0);
    parked_inst : out std_logic;
    granted_inst : out std_logic;
    locked_inst : out std_logic;
    grant_inst : out std_logic_vector(inst_n-1 downto 0);
    grant_index_inst : out std_logic_vector(bit_width(inst_n)-1
      downto 0)
  );
end DW_arb_sp_inst;
```

architecture inst of DW_arb_sp_inst is

begin

```
-- Instance of DW_arb_sp
U1 : DW_arb_sp
  generic map (
    n => inst_n,
    park_mode => inst_park_mode,
    park_index => inst_park_index,
    output_mode => inst_output_mode
  )
  port map (
    clk => inst_clk,
    rst_n => inst_rst_n,
    init_n => inst_init_n,
    enable => inst_enable,
```

```
        request => inst_request,
        lock => inst_lock,
        mask => inst_mask,
        parked => parked_inst,
        granted => granted_inst,
        locked => locked_inst,
        grant => grant_inst,
        grant_index => grant_index_inst
    );

end inst;

-- pragma translate_off
configuration DW_arb_sp_inst_cfg_inst of DW_arb_sp_inst is
    for inst
    end for; -- inst
end DW_arb_sp_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_arb_sp_inst( inst_clk, inst_rst_n, inst_init_n, inst_enable, inst_request,
                      inst_lock, inst_mask, parked_inst, granted_inst, locked_inst,
                      grant_inst, grant_index_inst );

parameter inst_n = 4;
parameter inst_park_mode = 1;
parameter inst_park_index = 0;
parameter inst_output_mode = 1;

`define bit_width_n 2// bit_width_n is set to ceil(log2(n))

input inst_clk;
input inst_rst_n;
input inst_init_n;
input inst_enable;
input [inst_n-1 : 0] inst_request;
input [inst_n-1 : 0] inst_lock;
input [inst_n-1 : 0] inst_mask;
output parked_inst;
output granted_inst;
output locked_inst;
output [inst_n-1 : 0] grant_inst;
output [`bit_width_n-1 : 0] grant_index_inst;

// Instance of DW_arb_sp
DW_arb_sp #(inst_n, inst_park_mode, inst_park_index, inst_output_mode) U1 (
    .clk(inst_clk),
    .rst_n(inst_rst_n),
    .init_n(inst_init_n),
    .enable(inst_enable),
    .request(inst_request),
    .lock(inst_lock),
    .mask(inst_mask),
    .parked(parked_inst),
    .granted(granted_inst),
    .locked(locked_inst),
    .grant(grant_inst),
    .grant_index(grant_index_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted content and title of “ Suppressing Warning Messages During Verilog Simulation ” on page 5 and added the DW_SUPPRESS_WARN macro
October 2019	DWBB_201903.5	■ Added the “Disabling Clock Monitor Messages” section
March 2019	DWBB_201903.0	■ Clarified license requirements in Table 1-3 on page 2 ■ Added “ Enabling minPower ” on page 7 ■ Added this Revision History table and the document links on this page

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